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momentum of a point mass is defined as the moment of its linear momentum. In other words, the angular momentum L of a point mass having a linear momentum p at a position r with respect to a point or axis is mathematically written as,

$$\vec{L} = \vec{r} \times \vec{p} \qquad (5.18)$$

The magnitude of angular momentum could be written as,

$$L = r\,p\sin\theta \qquad (5.19)$$

where, θ is the angle between $\vec{r}$ and $\vec{p}$. $\vec{L}$ is perpendicular to the plane containing $\vec{r}$ and $\vec{p}$. As we have written in the case of torque, here also we can associate $\sin\theta$ with either $\vec{r}$ or $\vec{p}$.

$$L = r\left(p\sin\theta\right) = r\left(p\perp\right) \qquad (5.20)$$

$$L = \left(r\sin\theta\right)p = \left(r\perp\right)p \qquad (5.21)$$

where, $p\perp$ is the component of linear momentum p perpendicular to r, and $r\perp$ is the component of position r perpendicular to p.

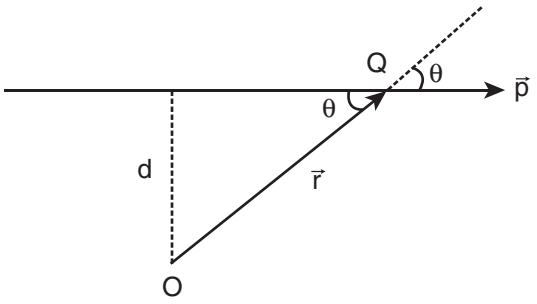
The angular momentum is zero $\left(L = 0\right)$, if the linear momentum is zero $(p = 0)$ or if the particle is at the origin $\left(\vec{r} = 0\right)$ or if $\vec{r}$ and $\vec{p}$ are parallel or antiparallel to each other ($\theta = 0^0$ or 180^0).

There is a misconception that the angular momentum is a quantity that is associated only with rotational motion. It is not true. The angular momentum is also associated with bodies in the linear motion. Let us understand the same with the following example.

EXAMPLE 5.11

A particle of mass (m) is moving with constant velocity (v). Show that its angular momentum about any point remains constant throughout the motion.

Solution



Let the particle of mass m move with constant velocity $\vec{v}$. As it is moving with constant velocity, its path is a straight line. Its momentum $\left(\vec{p} = m\vec{v}\right)$ is also directed along the same path. Let us fix an origin (O) at a perpendicular distance (d) from the path. At a particular instant, we can connect the particle which is at positon Q with a position vector $\left(\vec{r} = \overrightarrow{OQ}\right)$.

Take, the angle between the $\vec{r}$ and $\vec{p}$ as θ. The magnitude of angular momentum of that particle at that instant is,

$$L = OQ\,p\sin\theta = OQ\,mv\sin\theta = mv\left(OQ\sin\theta\right)$$

The term $\left(OQ\sin\theta\right)$ is the perpendicular distance (d) between the origin and line along which the mass is moving. Hence, the angular momentum of the particle about the origin is,

$$L = mvd$$

The above expression for angular momentum L, does not have the angle θ. As the momentum (p = mv) and the